Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer pursuing an MSCS at USC with 2+ years building **Java/C++/Python systems**, simulation-oriented tooling, and Linux-based workflows, with hands-on experience in debugging, CI/CD, and support for complex real-world software environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Retrieval, Web Technologies

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **platform services and validation workflows** for 500+ users, reducing request latency from **1.2s to 350ms** while improving issue diagnosis, operational visibility, and responsiveness across production software systems.
- Implemented **automated validation and ETL checks** for 5,000+ monthly jobs, reducing downstream failures while improving monitoring, release confidence, and production support readiness across recurring software workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **large-scale simulation pipelines** across 100,000+ trajectory records, enabling reproducible algorithm evaluation, faster diagnostics, and high-confidence benchmarking while improving signal quality for autonomy-related workflows.
- Produced **technical reports and test scenarios** translating findings into actionable guidance, improving cross-team alignment, reducing handoff ambiguity, and strengthening engineering quality as prototypes matured toward production readiness.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting distributed integrations, cross-platform validation, and stable delivery across production workloads.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and strengthening release quality for coordinated software components.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **backend features and APIs** handling 10,000+ daily requests, translating product requirements into maintainable services while improving supportability, implementation ownership, and production debugging workflows.
- Built **AI-assisted grading workflows** with multimodal models and OCR, reducing feedback turnaround by **60%** while supporting production operations through concurrent execution, CI/CD-backed releases, and cost-aware debugging.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **MERN and WebSocket modules** for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening responsiveness for real-time user-facing applications.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation and troubleshooting.

Skills

Languages: Java, C++, Python, JavaScript/TypeScript, SQL, Bash
Systems & Simulation: Linux, GCC/Make familiarity, simulation pipelines, debugging, cross-platform validation
Backend & Tooling: REST APIs, Flask, FastAPI, Jenkins, GitHub, CI/CD, automation
Networking & Platforms: TCP/IP, JSON, YAML, Windows/Linux development, repository workflows
Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, production support workflows
Engineering: unit testing, integration testing, technical documentation, code reviews, Agile/Scrum

Projects

CPU-scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** algorithms in a **C++**-backed simulator, pairing interactive controls with Gantt visualization across 1K+ process combinations.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring and schema exploration through a unified **Java** and **TypeScript** codebase with Docker-backed local workflows.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability, reducing deployment variance, and streamlining troubleshooting across production workflows.